

Research Highlights

Expert Systems with Applications (ESWA)

Manuscript Title: Evidential Multi-Agent Orchestration for Multimodal Urban Flood Risk Assessment: A Subjective Logic Framework with Provenance-Aware Explanations

- First framework coupling Subjective Logic algebra with multi-agent flood AI.
- Partial projection guarantees monotonic uncertainty under missing modalities.
- Brier-score credibility tracking prevents uncalibrated agent contributions.
- Prevents minority-class collapse, outperforming LLM arbitration on floods.
- DuckDB provenance ledger powers interactive per-cell explainability.